Technical Document - "We're Going to War"
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Algorithms and Calculations

The Cards
Two loops go through every suit and value, creating a card for each combination and adding it to the deck. This creates 52 cards, 13 cards for each of the 4 suits. Since the game has three players, one random card is removed using Math.random() to select a position and deck.splice() to remove it. This leaves 51 cards, allowing each player to receive 17 cards. The remaining cards are then shuffled using the Fisher-Yates Algorithm, which randomly swaps their positions to create a different deck order each game.

The Game
Using arrays, each player has a hand containing all of their cards. After the cards are all distributed, pressing n prompts a card to be selected from each player's hand and is placed on the table and faced up. Each card contains its own value, and the player that placed the highest ranking card wins the pot, and all obtains all 3 cards on the table. The first player to have zero cards left is out, and the game continues until only one player is left with all of the cards. If two or more of the players have a card with the same value, a war begins, and each player must place an additional card faced down and a card faced up. Whoever has the highest value among the faced up cards wins the entire pot.

The Bee
The Bee acts as the suspended light above the game table, and can be controlled around using WASD. I wanted to do an outdoor setting for the project, and the original plan was to just have the wire holding up the light go really far into the sky, but then I wanted to model some more, so I changed it to be a bee with a little ribbon tied around it to hold the light. The wings are even animated.

The Skybox
The skybox is a large cube that encases the setting, where it has a gradient from a dark blue to a lighter blue, with green near the bottom to match the grassy plane the table sits on (due to camera limitations, this isn't too visible). For detail, it has cloud textures along the sides, as long as the required Sun is positioned in the sky off to the right of the table.

Textures
The cards all have a different texture based on what playing card they are, the cards are textured with their suit and number values on the front, and all have a standard card back texture on the back. The face cards use imported images layered on top of the texture.

Challenges We Faced

Timing
The project had a decent amount of work to complete we didn't really take into consideration, so we were quite behind on some parts.

GitHub
GitHub went well, just forgot to update the KanBan board as we worked through the project.

Debugging
Trying to implement the various animations for the card game was a challenge, at one point, the card animation was under the table rather than on the surface. It took considerable time to tweak and fix these issues